SSC-YOLO: Spectral-Spatial Fusion Convolution for Ship Detection in SAR Imagery

Dung Hoang-Ngoc, Dat Minh-Tien Nguyen and Thien Huynh-The
Department of Computer and Communications Engineering
HCMC University of Technology and Education, Ho Chi Minh City, 71307, Vietnam
Email: dunghoang0401237@gmail.com, nguyenmtdat0501@gmail.com, thienht@hcmute.edu.vn

2025 International Conference on Advanced Technologies for Communications (ATC) | 979-8-3315-8698-0/25/$31.00 ©2025 IEEE | DOI: 10.1109/ATC67618.2025.11268806

***Abstract*—Object detection in synthetic aperture radar (SAR) imagery plays a vital role in maritime surveillance, national defense, and environmental monitoring. However, this task remains challenging due to speckle noise, low target contrast, and complex background clutter. To address these issues, we propose SSC-YOLO, a lightweight and efficient one-stage detector specifically designed for SAR-based ship detection, built upon the YOLOv10 architecture. The model introduces two key innovations: the Spectral-Spatial Convolution (SSC) module, which fuses spatial and Spectral-domain features using Fractional Gabor Filters (FGF) to enhance directional sensitivity; and the Multi-Scale Feature Aggregation (MSFA) module, which uses a dual-head structure in the neck to adaptively integrate features across scales, improving the detection of small objects without increasing inference cost. SSC-YOLO is evaluated on two public SAR datasets, SSDD and HRSID, achieving 98.8% $\mathbf{mAP_{0.5}}$ and 72.0% $\mathbf{mAP_{0.95}}$ on SSDD, as well as 89.5% $\mathbf{mAP_{0.5}}$ and 62.9% $\mathbf{mAP_{0.95}}$ on HRSID. It also maintains the smallest model size among all compared methods, with only 1.8 million parameters. Ablation studies confirm that each proposed module contributes effectively to performance enhancement. With high accuracy, compact design, and real-time capability, SSC-YOLO offers a practical solution for SAR object detection on platforms with limited computational resources, such as UAVs and embedded systems.**

***Index Terms*—ship detection, remote sensing, YOLO, small object, feature fusion.**

I. Introduction

Synthetic aperture radar (SAR) is an active microwave system that captures high-resolution Earth images regardless of light or weather. By simulating a large aperture via antenna-target motion, it achieves fine spatial detail without a large physical antenna. Its sensitivity to the backscattering properties of targets makes SAR a vital tool for marine surveillance, disaster response, resource exploration, and defense [1]. Despite these advantages, SAR imagery is inherently degraded by speckle noise and often contains numerous small, low-contrast targets that blend into complex backgrounds. Classical algorithms based on thresholding or handcrafted feature engineering struggle with variable scene contexts and multi-scale target detection [2]. In response, deep learning approaches have emerged to perform end-to-end feature extraction and classification, substantially improving detection accuracy.

Recently, research on object detection in SAR imagery has followed two main paradigms: one-stage and two-stage frameworks. One-stage methods, such as the YOLO [3] families and RetinaNet [4], perform detection in a single pass, which offers high inference speed but can struggle with small targets and remains vulnerable to speckle noise. By contrast, two-stage approaches exemplified by R-CNN [5], Fast R-CNN [6], and Faster R-CNN [7] first generate region proposals before carrying out detailed classification. This yields higher accuracy for multi-scale objects in complex scenes but incurs greater computational overhead and latency, making real-time operation more difficult. Similarly, anchor-free methods, while offering flexibility, often suffer from increased computational cost. For example, the two-stage detector proposed by Yue *et al.* [8] effectively enhances orientation sensitivity using high-resolution pyramids and dynamic rotated sampling, but its complex architecture significantly slows down inference. Similarly, NPA2Net [9] introduces nested path aggregation and attention to refine feature extraction, yet its reliance on heavy attention mechanisms leads to prolonged inference time, unsuitable for lightweight deployment. Tang *et al.* [10] address the challenging task of detecting small and near-shore ships by integrating deformable convolutions and BiFormer attention in DBW-YOLO. However, the complex architecture results in high computational costs, making deployment challenging on resource-constrained platforms. YOLO-OSD [11] enhances small ship detection through refined anchor design, yet its reliance on static anchors and the absence of oriented bounding box (OBB) modeling limit its robustness when handling rotated or arbitrarily oriented targets. Similarly, SSPNet [12] augments Faster R-CNN with attention and scale-aware modules to improve adaptability to varying object sizes. GMDR-Net [13], while supporting OBB detection through a lightweight architecture, introduces training instability due to the complexity of its Gaussian mixture-based enhancement and specialized loss functions.

Unlike natural images, SAR data relies heavily on pectral-domain features to distinguish targets from complex backgrounds. Reducing speckle noise enhances image clarity and analytical reliability, thereby improving its applicability in remote sensing [14]. Inspired by the limitations of prior approaches and recent advances in one-stage object detectors, we adopt YOLOv10 [15] as the baseline architecture due to its well-documented balance between accuracy and computational efficiency. Building on this foundation, we propose a novel framework to address challenges in ship detection, clutter suppression, and pectral feature representation in SAR imagery, with two key contributions:

Corresponding author: Thien Huynh-The (thienht@hcmute.edu.vn)

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.

979-8-3315-8698-0/25/$31.00 ©2025 IEEE

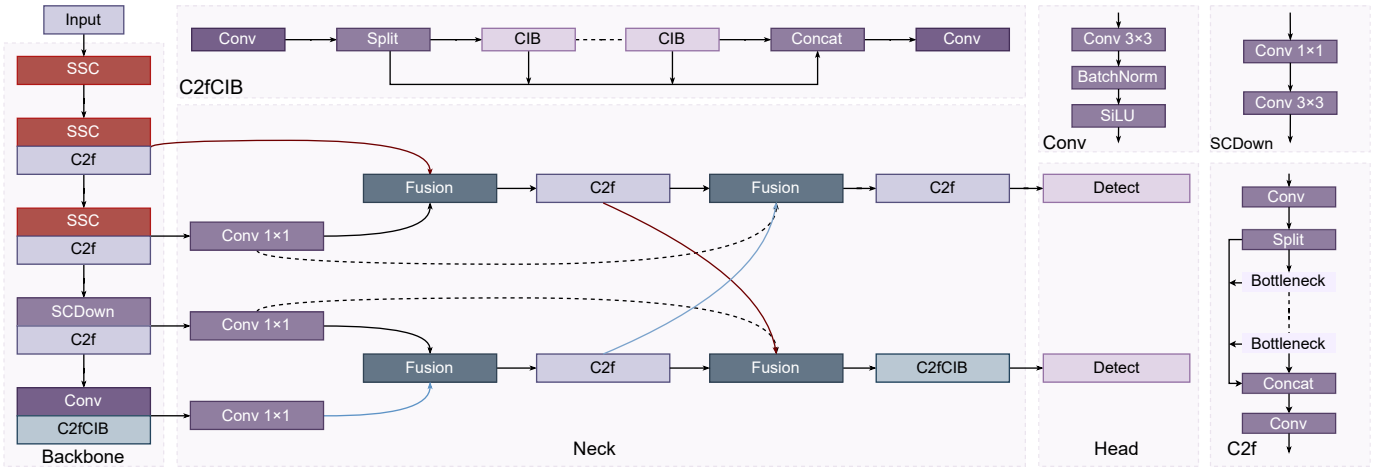


Fig. 1. Overview of the SSC-YOLO architecture. The model integrates enhanced YOLOv10 components (C2f, C2fCIB, SCDown) with two novel modules: SSC for spectral-spatial feature fusion and MSFA for adaptive multi-scale aggregation. Blue arrows indicate upsampling, while red arrows denote downsampling between feature levels.

- **Spectral-Spatial Convolution:** We design a novel convolutional block that jointly captures spatial and Spectral-domain features using Fractional Gabor Filters. This module enhances feature diversity and discriminability while effectively reducing inter-channel redundancy, thereby improving representation learning under the challenging conditions of SAR imagery.
- **Multi-Scale Feature Aggregation:** We propose a lightweight and resolution-aware neck design that adaptively aggregates multi-scale features from different levels of the backbone. This structure enables more effective detection of small and multi-scale targets by reinforcing semantic coherence across scales while maintaining real-time inference capability.

II. Methodology

The overall architecture of our proposed model, named SSC-YOLO, is illustrated in Fig. 1. Building upon the YOLOv10 backbone, our design introduces novel components to enhance feature representation and improve detection performance, particularly in complex SAR imagery.

A. Spectral-Spatial Convolution

In this section, the proposed Spectral-Spatial Fusion (SSC) module is introduced, as depicted in Fig. 2, which is composed of three main components: depthwise separable convolution (DSC), a spatial processing domain (SPD), and a spectral feature extractor (FED). Initially, the input is downsampled via DSC to enhance computational efficiency. Subsequently, the architecture bifurcates into two parallel branches. The first branch is responsible for enhancing global semantic representation. Meanwhile, the second branch emphasizes diverse features in spatial and pectral dimensions and further selects more representative features, contributing to improved.

1) Depthwise Separable Convolution: Let $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$ represent the input tensor. The DSC first applies a depthwise convolution (DWC) [16] for channel-wise spatial filtering, followed by a pointwise convolution (PWC) [17] to fuse inter-channel information. This decoupled structure reduces computation while preserving representational power. The resulting feature maps are then normalized using batch normalization (BN) and activated by ACON-C [18]. The ACON-C introduces a learnable and adaptive non-linearity, formulated as:

$$\text{ACON-C}(\mathbf{X}) = (\mathbf{P}_1-\mathbf{P}_2)\odot\mathbf{X}\odot\sigma\big(\mathbf{B}(\mathbf{P}_1-\mathbf{P}_2)\odot\mathbf{X}\big)+\mathbf{P}_2\odot\mathbf{X}, \tag{1}$$

where $\mathbf{P}_1, \mathbf{P}_2 \in \mathbb{R}^{1\times1\times C}$ adaptive gating parameters and β are learnable parameters, $\sigma(\cdot)$ denotes the sigmoid activation function, and $\odot$ represents element-wise multiplication. Specifically, $\mathbf{P}_1 - \mathbf{P}_2$ serves as a learnable scaling factor that modulates the non-linearity's strength, while $\mathbf{P}_2$ enables residual scaling, allowing identity mapping when necessary. Finally, the computational process of DSC with a kernel size of $n \times n$ can be compactly expressed as follows:

$$\text{DSC}_{n\times n}(\mathbf{X}) = \text{ACON-C}\left(\text{BN}\left(\text{PWC}\left(\text{DWC}(\mathbf{X})\right)\right)\right) \tag{2}$$

2) Spatial Processing Domain: For spatial analysis, SPD is designed to enrich contextual features while ensuring computational efficiency. In the detailed architecture presented in Fig. 3(a), the input feature map $\mathbf{X}^s \in \mathbb{R}^{H\times W\times C}$ is evenly partitioned into two distinct channel groups, each belonging to $\mathbb{R}^{H\times W\times \frac{C}{2}}$, such that $\mathbf{X}^s = [\mathbf{X}_1, \mathbf{X}_2]$. The dual-branch structure captures multi-scale spatial features efficiently:

$$\begin{cases}\mathbf{F}_1 = \mathcal{G}_{3\times3}(\mathbf{X}_1),\\ \mathbf{F}_2 = \mathcal{G}_{5\times5}(\mathbf{X}_2+\mathbf{F}_1),\end{cases} \tag{3}$$

where $\mathcal{G}_{3\times3}$ and $\mathcal{G}_{5\times5}$ denote grouped convolution operations with kernel sizes 3×3 and 5×5, respectively. The resulting feature map $\mathbf{F}_1, \mathbf{F}_2 \in \mathbb{R}^{H\times W\times \frac{C}{2}}$ are concatenated along the channel axis:

$$\mathbf{F}_{\text{cat}} = [\mathbf{F}_1, \mathbf{F}_2] \in \mathbb{R}^{H\times W\times C}. \tag{4}$$

The concatenated feature map is then passed through a PWC to perform channel mixing and obtain the final output feature map $\mathbf{Y}^s \in \mathbb{R}^{H\times W\times C}$:

$$\mathbf{Y}^s = \text{PWC}(\mathbf{F}_{\text{cat}}). \tag{5}$$

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.

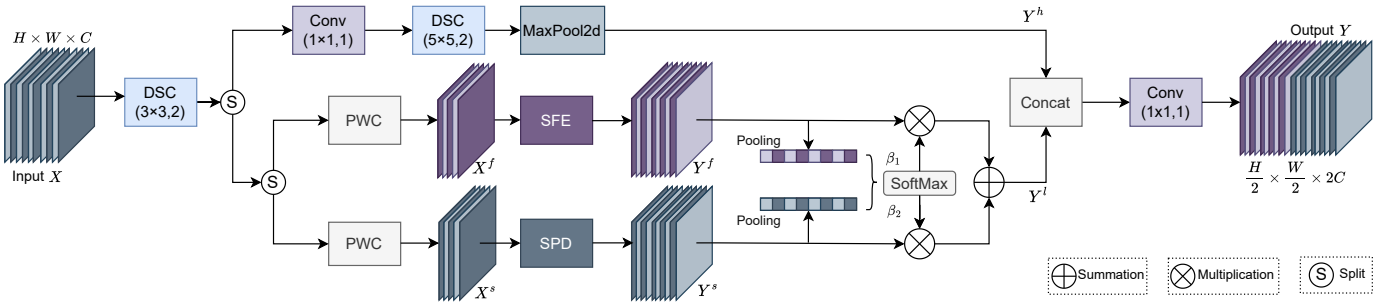


Fig. 2. The SSC module integrates a lightweight downsampling convolution with two parallel branches. The first branch focuses on enhancing spatial feature representations, while the second leverages an SPD for local spatial feature extraction and SFE employing FGF for pectral-domain representation. Following channel-wise splitting and adaptive weighting, the features are fused and refined through a 1×1 convolution, effectively capturing complementary spatial and spectral information to enhance representational capacity.

3) Spectral Feature Extractor: In SAR imagery with complex textures and noise. Traditional Gabor Filters can be embedded into convolutional layers to improve robustness against geometric transformations such as scaling, rotation, and deformation [19]. The FGF extends the classical Gabor transform by introducing a fractional order parameter, enabling more flexible time-pectral localization [20]. This allows finer modeling of spatial- spectral features under transformations like rotation and scaling, as illustrated in Fig. 3(b). For a given location (x, y), the normalized spatial coordinates in the range $[-1, 1]^2$ are rotated and scaled based on a target orientation θ and scale factor s, as defined:

$$\begin{aligned} x_\theta &= x\cos\theta + y\sin\theta, \\ y_\theta &= -x\sin\theta + y\cos\theta. \end{aligned} \tag{6}$$

CNN struggle to capture orientation and structural variations, while FGF enhances multiscale feature extraction using Gaussian-modulated cosine wavelets. Based on the transformed coordinates, the real-valued FGF $G_s^{(\alpha)} \in \mathbb{R}^{k\times k}$ is constructed as follows:

$$G_s^\alpha(x_\theta, y_\theta) = \exp\left(-\left(x_\theta^2 + \left(\frac{y_\theta}{s}\right)^2\right)^\alpha\right) \cdot \cos\left(\frac{2\pi x_\theta}{s}\right), \tag{7}$$

where s is a scale parameter s and fractional order α controlling the sharpness of the kernel, and (x_θ, y_θ) represent the rotated coordinates of the spatial point (x, y) with respect to the angle θ. Accordingly, a collection of such kernels $G_s^{(\alpha)}$ is generated for multiple orientations and scales. These kernels are integrated with learnable convolutional weights $\mathbf{T} \in \mathbb{R}^{C\times k\times k}$ as described below:

$$\mathbf{W}_\mathrm{G} = \sum_{i=1}^{N} G_{s_i}^\alpha(x_\theta, y_\theta) \odot \mathbf{T}, \tag{8}$$

where $N = |\Theta| \cdot |\mathcal{S}|$ represents the total number of FGF generated over all predefined orientation and scale combinations. In this study, the orientation set is defined as $\Theta = \{0°, 45°, 90°, 135°\}$, and the scale set as $\mathcal{S} = \{1, 2, 3, 4\}$. These Gabor-enhanced representations are subsequently processed to enhance feature selectivity and robustness. Specifically, the resulting output is passed through a non-linear activation, dropout, then a Squeeze-and-Excitation (SE) attention module [21] that adaptively recalibrates channel-wise features:

$$\mathbf{A} = \sigma\left(\mathrm{Fc}_2\left(\delta\left(\mathrm{Fc}_1\left(\mathrm{GAP}(\mathbf{W}_\mathrm{G})\right)\right)\right)\right), \tag{9}$$

where $\mathrm{GAP}(\cdot)$ denotes the global average pooling operation, $\delta(\cdot)$ is the SiLU non-linear activation function [22], and $\sigma(\cdot)$ represents the sigmoid function. The two fully connected transformations, implemented as learnable 1×1 convolution layers, are defined as $\mathrm{Fc}_1 : \mathbb{R}^C \to \mathbb{R}^{C/r}$ and $\mathrm{Fc}_2 : \mathbb{R}^{C/r} \to \mathbb{R}^C$, where r is the reduction ratio. These layers first compress and then restore the channel dimensionality. The resulting attention vector $\mathbf{A} \in \mathbb{R}^{1\times1\times C}$ is subsequently applied to the input feature map $\mathbf{W}_\mathrm{G} \in \mathbb{R}^{H\times W\times C}$, producing an attention-weighted feature representation, computed as:

$$\mathbf{Y}^f = \mathcal{C}_{1\times1}(\mathbf{W}_\mathrm{G} \odot \mathbf{A}), \tag{10}$$

where $\mathcal{C}_{1\times1}(\cdot)$ denotes a channel-wise transformation implemented using a 1×1 convolution.

4) Spetral-Spatial Fusion Convolution: : Let $\mathbf{Y}^h, \mathbf{Y}^l$ denote the output from the first branch and second branch. In the first brand, global contextual features are enhanced by applying a PWC followed by a DSC with a 5×5, effectively expanding the receptive field to capture broad contextual information, and finally refined through a pooling as follows:

$$\mathbf{Y}^h = \mathrm{GMP}\left(\mathrm{DSC}_{5\times5}\left(\mathrm{PWC}(\mathrm{DSC}_{3\times3}(\mathbf{X}))\right)\right), \tag{11}$$

where $\mathbf{X} \in \mathbb{R}^{H\times W\times\frac{C}{2}}$ denotes the input feature map, and GMP refers to global max pooling. In the second branch, we integrate both pectral and spatial information by applying global average pooling, as formulated:

$$\mathbf{P^n} = \mathrm{GAP}(\mathbf{Y}^n) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\mathbf{Y}_{i,j}^n, \quad n \in \{s, f\}. \tag{12}$$

Next, we stack the upper and lower global channel descriptors $\mathbf{P^s}, \mathbf{P^f} \in \mathbb{R}^{1\times1\times C}$, and compute soft attention weights using a channel-wise softmax function:

$$\beta_1 = \frac{e^{\mathbf{P^s}}}{e^{\mathbf{P^s}} + e^{\mathbf{P^f}}}, \quad \beta_2 = \frac{e^{\mathbf{P^f}}}{e^{\mathbf{P^s}} + e^{\mathbf{P^f}}}, \tag{13}$$

where $\beta_1, \beta_2 \in \mathbb{R}^C$ with element-wise exponentials. The output of the second branch is computed as a weighted sum of the two feature maps, as formulated:

$$\mathbf{Y}^l = \beta_1 \odot \mathbf{Y}^s + \beta_2 \odot \mathbf{Y}^f. \tag{14}$$

The final output $\mathbf{Y} \in \mathbb{R}^{\frac{H}{2}\times\frac{W}{2}\times2C}$ is concated $\mathbf{Y}^h$ with $\mathbf{Y}^l$ followed by a convolutional transformation:

$$\mathbf{Y} = \mathcal{C}_{1\times1}([\mathbf{Y}^h, \mathbf{Y}^l]). \tag{15}$$

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.

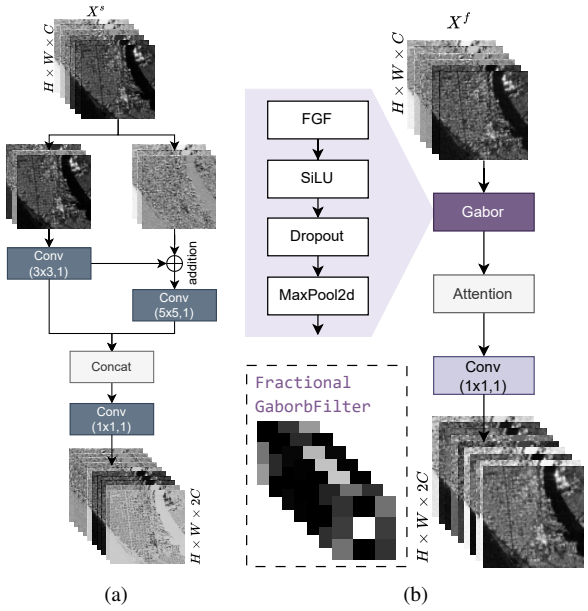


Fig. 3. (a) The detail architecture of SPD splits the input feature map into two halves processed by depthwise convolutions, then fuses outputs with a 1×1 convolution and residual connection to capture multi-scale spatial features efficiently. (b) The detail of SFE enhances feature representation by applying FGF with SiLU activation and attention mechanisms.

This unified mechanism adaptively integrates complementary spatial and frequency cues, enabling more discriminative and robust feature learning.

B. Multi-Scale Feature Aggregation

To effectively detect small scale ships in SAR imagery under limited computational resources, SSC-YOLO integrates a lightweight yet expressive Multi-Scale Feature Aggregation (MSFA) module in its neck design. The MSFA is tightly coupled with an SSC-enhanced backbone and a refined dual-head output strategy, as illustrated in Fig. 1. The backbone incorporates three stacked SSC modules, which combine spatial-domain convolution with spectral-domain modeling via FGFs. These modules enhance low-level feature extraction by capturing orientation-sensitive patterns and mitigating speckle noise in SAR images. Compared to traditional convolutional layers, the SSC modules improve local semantic richness and yield more robust representations for downstream processing.

Following the SSC blocks, three distinct 1×1 convolution layers are introduced at different backbone depths. These layers serve as channel refiners and alignment units, harmonizing feature dimensions before aggregation. Let the output feature maps from the backbone be denoted as: $\mathbf{F}_3 \in \mathbb{R}^{160\times160\times C_3}$, $\mathbf{F}_4 \in \mathbb{R}^{80\times80\times C_4}$, $\mathbf{F}_5 \in \mathbb{R}^{40\times40\times C_5}$. These feature maps represent high, medium, and low spatial resolutions, respectively. To integrate them, MSFA employs a bidirectional and weighted fusion mechanism. For each fusion level l, the aggregated feature map is computed as:

$$\mathbf{F}_{\text{agg}}^{(l)} = \frac{\sum_i w_i^{(l)} \cdot \mathcal{R}(\mathbf{F}_i)}{\epsilon + \sum_j w_j^{(l)}},$$

where $w_i^{(l)} \geq 0$ are trainable non-negative weights enforced via ReLU activation, $\mathcal{R}(\cdot)$ denotes resolution alignment via upsampling or downsampling, and $\epsilon = 10^{-4}$ ensures numerical stability. In addition to vertical fusion, MSFA introduces cross-level shortcut connections to facilitate richer information flow. The outputs from higher-resolution branches are routed to deeper fusion paths. These diagonal connections preserve spatial granularity and compensate for semantic gaps introduced by downsampling. This is particularly beneficial for detecting small and low-contrast targets commonly encountered in SAR datasets. The aggregated features are refined via C2f and C2fCIB modules before being passed to two detection heads at 160×160 and 80×80 resolutions, respectively. The lowest resolution head 40×40 is intentionally omitted to reduce model complexity. This dual-head configuration balances precision and efficiency: the higher-resolution head targets small object localization, while the lower-resolution head captures contextual semantics.

III. EXPERIMENTAL RESULTS

This section presents a systematic evaluation conducted on two benchmark SAR datasets: SSDD [23] and HRSID [24]. First, a brief overview of each dataset is provided, followed by an in-depth explanation of the SSC-YOLO architecture. To ensure applicability in resource-constrained environments, such as embedded systems or unmanned aerial vehicles, the proposed model is developed based on YOLOv10n the most lightweight variant of the YOLOv10 series and is adopted as the baseline for all subsequent experiments. Ablation studies are then conducted on the SSDD dataset to assess the contribution of each enhancement module by progressively removing them from SSC-YOLO. Lastly, a comparative analysis is performed against existing the state-of-the-art methods, evaluating both detection accuracy and inference speed across SSDD and HRSID. This comprehensive evaluation demonstrates the effectiveness, efficiency, and deployment potential of SSC-YOLO in real-world SAR object detection applications.

A. Dataset Description

SSDD: The SSDD dataset comprises $1,160$ SAR images of various dimensions, containing a total of $2,456$ ship instances captured under diverse operational conditions, including different resolutions, vessel scales, and sea states. Its environmental and scale diversity makes SSDD a comprehensive benchmark for evaluating ship detection algorithms in SAR imagery.

HRSID: The HRSID dataset includes $5,604$ high resolution SAR images, each resized to 800×800 pixels, with spatial resolutions ranging from 1m to 5m. It contains $16,951$ annotated ship instances. Unlike SSDD, HRSID supports polygon-based annotations, facilitating precise instance level segmentation by accurately capturing ship contours and geometric details.

B. Training Setups and Evaluation Metrics

All experiments were carried out on a setup featuring an Intel i7-9700 CPU, 16GB of RAM, and an NVIDIA GTX 2080 GPU. Each comparison model was trained for up to

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.

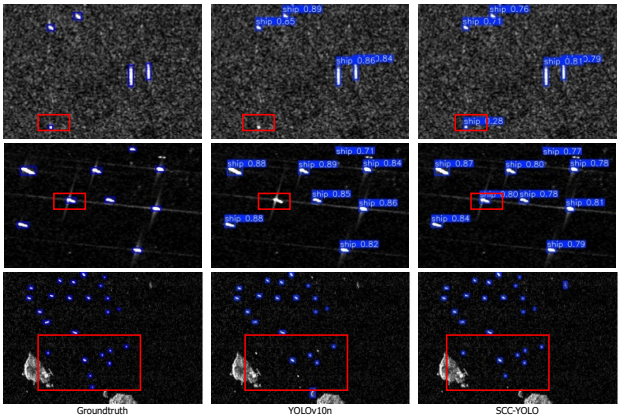


Fig. 4. Visualization of detection results from YOLOv10n and SSC-YOLO on SAR ship images from SSDD and HRSID. Red bounding boxes indicate SSC-YOLO's enhanced ability to detect small and low-contrast vessels under cluttered and noisy conditions, demonstrating superior robustness.

TABLE I
OUR ABLATION STUDY OF SSC-YOLO WITH KEY COMPONENTS (A: SSC MODULE AND B: MSFA-) ON THE SSDD DATASET

Components		Pr	Re	$mAP_{0.5}$	$mAP_{0.95}$	Time	Params
A	B	(%)	(%)	(%)	(%)	(ms)	(M)
✗	✗	91.4	93.9	97.4	72.6	41.6	2.3
✗	✓	96.0	**96.2**	98.4	**73.2**	**35.7**	2.0
✓	✗	94.9	96.2	98.1	71.2	40.3	2.2
✓	✓	**96.8**	94.9	**98.8**	72.0	36.4	1.8

$1,000$ epochs, with early stopping triggered if no improvement was observed for 100 consecutive epochs. We maintained YOLOv10's default parameters to ensure consistency and reproducibility of results. For assessing model performance, we employed the mean average precision (mAP), a widely accepted metric that evaluates the balance between precision (Pr) and recall (Re) across various intersection over union (IoU) thresholds. In particular, $mAP_{0.5}$ and $mAP_{0.95}$ were utilized to gauge detection accuracy, providing a thorough insight into both localization precision and detection robustness.

C. Results and Discussions

1) Ablation Study: To evaluate the effectiveness of each proposed module, a comprehensive ablation study is conducted on the SSDD dataset, with results summarized in Table I. All configurations are built upon the YOLOv10n backbone to maintain consistent conditions and enable fair comparisons. The two primary components under investigation are the SSC and MSFA modules. Each variant includes or excludes these modules to assess their impact. Evaluation metrics include precision, recall, $mAP_{0.5}$, $mAP_{0.95}$, inference time, and model size. SSC-YOLO demonstrates a clear visual advantage over YOLOv10n low contrast vessels in noisy SAR imagery, thereby confirming its robustness and practical utility beyond quantitative results, as illustrated in Fig. 4.

The baseline model achieves $mAP_{0.5}$ of 97.4% and $mAP_{0.95}$ of 72.6%, along with a precision of 91.4% and a model size of 2.3M parameters. When the MSFA module is added alone, performance improves noticeably, with precision reaching 96.0%, $mAP_{0.5}$ rising to 98.4%, inference time reduced to 35.7 ms, and the parameter count lowered to 2.0M. These results indicate that MSFA significantly enhances multi-scale feature fusion while simplifying the architecture through its dual-head detection strategy. Introducing only the SSC module enhances the model's spatial-spectral representation, yielding a recall of 96.2% and $mAP_{0.5}$ of 98.1%. However, the inference time slightly increases to 40.3 ms due to the additional spectral-domain processing introduced by the FGFs. The SSC module proves especially useful in detecting small or low-contrast ships within noisy SAR imagery by capturing orientation-aware and scale-adaptive features. The full SSC-YOLO model configuration delivers the best overall results, with the highest precision at 96.8% and $mAP_{0.5}$ reaching 98.8%. It also offers the most compact model, with only 1.8M parameters. While $mAP_{0.95}$ slightly drops to 72.0% compared to the baseline, the combination of high accuracy, low latency, and lightweight architecture makes SSC-YOLO an ideal choice for real-time SAR ship detection on platforms with limited computational resources, such as UAVs or embedded systems.

TABLE II
METHOD COMPARISON BETWEEN SSC-YOLO AND OTHER DEEP MODELS.

SSDD			
Model	$mAP_{0.5}(\%)$	$mAP_{0.95}(\%)$	Params (M)
YOLOv10n [15]	97.4	**72.6**	2.3
SSPNet [12]	91.6	58.3	-
YOLO-OSD [11]	97.7	63.8	38.0
Yue *et al.* [8]	90.7	-	43.3
NPA2Net [9]	92.1	61.6	16.8
GMDR-Net [13]	87.6	-	3.4
SSC-YOLO (Our)	**98.8**	72.0	**1.8**
HRSID			
Model	$mAP_{0.5}(\%)$	$mAP_{0.95}(\%)$	Params (M)
YOLOv10n [15]	88.3	**63.1**	2.3
Yue *et al.* [8]	86.5	-	43.3
DBW-YOLO. [10]	88.9	-	-
NPA2Net [9]	87.0	55.9	16.8
GMDR-Net [13]	77.1	-	3.4
SSC-YOLO (Our)	**89.5**	62.9	**1.8**

2) Comparison Results: To comprehensively evaluate the performance and efficiency of our proposed SSC-YOLO model, we conduct benchmark experiments against several state-of-the-art deep object detectors. Table II presents quantitative comparisons on two standard SAR datasets: SSDD and HRSID, capturing both detection accuracy and model compactness. On the SSDD dataset, SSC-YOLO demonstrates leading detection performance with a top $mAP_{0.5}$ of 98.8% , outperforming all compared models, including YOLO-OSD 97.7% and YOLOv10n 97.4%. When evaluated under a stricter localization threshold, our model achieves a competitive $mAP_{0.95}$ of 72.0%, slightly lower than YOLOv10n 72.6% but significantly surpassing NPA2Net 61.6% and YOLO-OSD 63.8%. This consistent accuracy under both coarse and fine thresholds highlights the robustness of our SSC-YOLO model in complex maritime SAR environments. For the HRSID dataset, which contains more cluttered and small-scale targets, SSC-YOLO continues to exhibit superior performance. It achieves the highest $mAP_{0.5}$ of 89.5%, exceeding YOLOv10n 88.3% and DBW-YOLO 88.9%. Moreover, it attains an $mAP_{0.95}$ of 62.9%, closely matching YOLOv10n

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.

63.1% while outperforming NPA2Net 55.9% and GMDR-Net below 78% $mAP_{0.5}$. Despite its strong detection performance, we perform the fewest SSC-YOLO model remains remarkably lightweight, with only 1.8M parameters among all compared models. In contrast, YOLO-OSD and the method of Yue *et al* require 38M and 43.3M parameters, respectively. Even so-called lightweight detectors like GMDR-Net 3.4M and YOLOv10n 2.3M possess a larger model footprint. This efficiency allows SSC-YOLO to be readily deployable on resource-limited platforms such as embedded systems or real-time UAVs, without compromising accuracy. In summary, SSC-YOLO achieves a favorable balance between detection precision and model complexity. The integration of our SSC module and the MSFA module used in the neck contributes to its ability to capture rich spatial-pectral cues and adaptively combine multi-resolution features. Although slightly behind YOLOv10n in terms of $mAP_{0.95}$ on SSDD, our SSC-YOLO model consistently outperforms others in $mAP_{0.5}$ while maintaining the lowest parameter count. These findings underline its practicality and effectiveness for real-world SAR ship detection tasks, particularly in edge-computing scenarios.

IV. Conclusions

In this paper, we propose SSC-YOLO model, a lightweight and efficient one-stage object detector tailored for SAR-based ship detection. The framework is designed to address key challenges inherent in SAR imagery, such as small object sizes, low contrast, and cluttered backgrounds. Central to SSC-YOLO are two novel components: the SSC module, which captures both spatial and spectral features through structured submodules (DSC, SPD, and SFE), and the MSFA module, which enables adaptive multi-scale feature fusion via a dual-head strategy. These modules contribute to improving robustness against noise and reducing model redundancy. Comprehensive experiments on two benchmark SAR datasets, SSDD and HRSID, demonstrate the superior performance of SSC-YOLO. The model achieves 98.8% $mAP_{0.5}$ and 72.0% $mAP_{0.95}$ on SSDD, and 89.5% $mAP_{0.5}$ and 62.9% $mAP_{0.95}$ on HRSID, while maintaining the lowest model complexity with only 1.8 million parameters. Compared to existing state-of-the-art methods, SSC-YOLO offers a better trade-off between detection accuracy and computational efficiency. Ablation studies confirm the individual contributions of SSC and MSFA in performance gains. Future research explores enhancements in spectral feature encoding, adding multi-scale attention, and the use of neural architecture search to further refine the balance between accuracy and model complexity.

Acknowledgment

This study was supported by Ho Chi Minh City University of Technology and Education under Research Grant No. SV2025-54.

References

[1] G.-V. Nguyen and T. Huynh-The, "Enhancing aerial semantic segmentation with feature aggregation network for deeplabv3+," *IEEE Trans. Geosci. Remote. Sens.*, vol. 21, pp. 1–5, Jul. 2024.

[2] T. Huynh-The, S. N. Truong, and G.-V. Nguyen, "Hbsenet: A hybrid bilateral network for accurate semantic segmentation of remote sensing images," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 14 179–14 193, Aug. 2024.

[3] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE/CVF CVPR*, Jun. 2016, pp. 779–788.

[4] Y. Chen *et al.*, "Research on recognition of fly species based on improved retinanet and cbam," *IEEE Access*, vol. 8, pp. 102 907–102 919, May 2020.

[5] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proc. IEEE/CVF CVPR*, Jun. 2014, pp. 580–587.

[6] R. Girshick, "Fast r-cnn," in *Proc. IEEE/CVF ICCV*, Dec. 2015, pp. 1440–1448.

[7] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[8] T. Yue *et al.*, "A precise oriented ship detector in sar images based on dynamic rotated positive sample mining," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 16, pp. 10 022–10 035, Oct 2023.

[9] C. Zhang *et al.*, "Npa2net: A nested path aggregation attention network for oriented sar ship detection," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 9772–9789, Apr. 2024.

[10] X. Tang, J. Zhang, Y. Xia, and H. Xiao, "Dbw-yolo: A high-precision sar ship detection method for complex environments," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 7029–7039, Mar. 2024.

[11] M. F. Humayun *et al.*, "Yolo-osd: Optimized ship detection and localization in multiresolution sar satellite images using a hybrid data-model centric approach," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 5345–5363, Feb. 2024.

[12] Y. Gong *et al.*, "Small ship detection of sar images based on optimized feature pyramid and sample augmentation," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 16, pp. 7385–7392, Aug. 2023.

[13] Y. Guo *et al.*, "Gmdr-net: A lightweight obb-based sar ship detection model based on gaussian mixture data augmentation and distance rotation iou loss," *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, vol. 17, pp. 11 931–11 942, Jun. 2024.

[14] S. V. R, R. Mahadevan, B. L. S, M. Periasamy, R. C. Raman, and L. M, "Speckle noise analysis for synthetic aperture radar (sar) space data," 2024. [Online]. Available: https://arxiv.org/abs/2408.08774

[15] A. Wang *et al.*, "YOLOv10: Real-time end-to-end object detection," *arXiv preprint arXiv:2405.14458*, May 2024.

[16] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *Proc. IEEE/CVF CVPR*, Apr. 2017, pp. 1251–1258.

[17] B.-S. Hua, M.-K. Tran, and S.-K. Yeung, "Pointwise convolutional neural networks," in *Proc. IEEE/CVF CVPR*, Mar. 2018, pp. 984–993.

[18] N. Ma, X. Zhang, M. Liu, and J. Sun, "Activate or not: Learning customized activation," in *2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2021, pp. 8028–8038.

[19] S. Luan, C. Chen, B. Zhang, J. Han, and J. Liu, "Gabor convolutional networks," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4357–4366, May 2018.

[20] A. Akan and Y. Çekiç, "A fractional gabor expansion," *Journal of the Franklin Institute*, vol. 340, no. 5, pp. 391–397, Aug. 2003.

[21] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF CVPR*, Jun. 2018, pp. 7132–7141.

[22] S. Elfwing, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural networks*, vol. 107, pp. 3–11, Nov. 2018.

[23] J. Li, C. Qu, and J. Shao, "Ship detection in sar images based on an improved faster r-cnn," in *2017 SAR in Big Data Era: Models, Methods and Applications (BIGSARDATA)*, Nov. 2017, pp. 1–6.

[24] S. Wei *et al.*, "Hrsid: A high-resolution sar images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120 234–120 254, Jun. 2020.

Authorized licensed use limited to: onsemi. Downloaded on October 08,2026 at 02:50:16 UTC from IEEE Xplore. Restrictions apply.